

MID-POINT REPORT

Week 3 Deliverable — PCAP Forensics Analysis

Date: 10 May 2025 | Team: Muhammad Ahmed, Hamna Rizwan, Eshal Tanveer

1. Introduction

A financial data center reported suspicious network behaviour on June 13, 2025. The network team captured all traffic during the incident and saved it as a PCAP file for forensic investigation. Our team was assigned to act as network security analysts, examine the captured traffic, identify the nature of the attack, and recommend mitigation strategies.

The PCAP file analysed in this report is: 2025-06-13-traffic-analysis-exercise.pcap (42MB, 48,877 packets, captured over 34 minutes).

1.1 Objectives

- Identify the attacker IP address and victim IP address
- Determine the type of attack that occurred
- Find the exact timestamp when the attack began
- Provide evidence-based proof of the attack using Wireshark
- Recommend mitigation strategies to prevent future attacks

1.2 Methodology

The team followed a structured 5-phase analysis approach:

- Phase 1: Obtained and extracted the PCAP file from the university LMS
- Phase 2: Performed initial triage in Wireshark — checked packet count, protocols, and time range
- Phase 3: Applied targeted display filters to isolate suspicious traffic
- Phase 4: Classified the attack type based on traffic patterns and packet behaviour
- Phase 5: Documented all findings and developed mitigation recommendations

2. Findings

2.1 Capture Overview

Property	Value
PCAP Filename	2025-06-13-traffic-analysis-exercise.pcap
File Size	42 MB
Total Packets	48,877 packets
Capture Start Time	June 13, 2025 — 15:33:55 UTC
Capture Duration	34 minutes (2,072 seconds)
Primary Protocol	TCP (47,300 frames — 97% of all traffic)

2.2 Key Findings

Finding	Details
Victim IP Address	10.6.13.133
Attacker IP Address	83.137.149.15
Attack Type	DDoS Attack — TCP SYN Flood
Attack Port	Port 443 (HTTPS)
First Suspicious Packet	Packet 8567
Attack Start Time	June 13, 2025 — 15:36:33 UTC
Total Attack Packets	35,485 packets (73% of all captured traffic)
Attack Data Volume	33 MB sent by attacker vs 754 KB from victim

2.3 Evidence of Attack

Evidence 1 — Abnormal Packet Volume

The attacker IP 83.137.149.15 sent 35,485 packets to the victim in just 136 seconds. This equals approximately 260 packets per second. Normal traffic between two IPs would be a few hundred packets total. 260 packets per second clearly indicates a flood attack.

Evidence 2 — Extremely One-Sided Traffic

The attacker sent 33MB of data while the victim could only respond with 754KB. This means the attacker sent 44 times more data than the victim could handle — the defining characteristic of a DDoS amplification attack.

Evidence 3 — Targeted Port 443 Flood

All attack traffic was directed at port 443 — the HTTPS port used for secure web traffic. By flooding this port the attacker prevented any legitimate HTTPS connections to the victim server, taking the web service offline.

3. Mitigation Recommendations

The following measures would have prevented or significantly reduced the impact of this attack:

3.1 Rate Limiting

Configure the firewall to automatically block any IP address that sends more than 100 packets per second. Since the attacker was sending 260 packets per second, this rule would have blocked the attack within the first second. Rate limiting is a simple and effective first line of defence against flood-based DDoS attacks.

3.2 IP Blacklisting

Permanently block the attacker IP address 83.137.149.15 at the firewall level. Once an attack is detected from a specific IP, that IP should be immediately added to a blocklist so no further traffic from that address can reach the network.

3.3 DDoS Protection Service

Deploy a cloud-based DDoS protection service such as Cloudflare or AWS Shield. These services sit between the internet and the company network and automatically detect and absorb DDoS traffic before it reaches the target server.

3.4 SYN Cookies

Enable SYN cookies on the victim server. SYN cookies verify whether each connection request is legitimate before allocating server resources. This prevents the server from being overwhelmed by thousands of fake connection requests during a SYN flood attack.

3.5 Intrusion Detection System

Install a network IDS such as Snort or Suricata. These systems monitor live traffic and automatically alert administrators when attack patterns are detected — such as a sudden spike in SYN packets from a single IP — allowing early response before damage occurs.

4. Conclusion

The forensic analysis of the PCAP file confirmed that the financial data center was targeted by a DDoS attack on June 13, 2025 at 15:36:33 UTC. The attacker at IP address 83.137.149.15 flooded the victim at 10.6.13.133 with 35,485 packets over 136 seconds, targeting port 443 to disrupt HTTPS services.

The evidence clearly shows an abnormal volume of one-sided traffic consistent with a TCP SYN Flood DDoS attack. The five recommended mitigations — rate limiting, IP blacklisting, DDoS protection services, SYN cookies, and an IDS — would collectively prevent this type of attack from succeeding in the future.

